ISYE 671: Integer Programming Exercises (Write-up)

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Problem 1: LP Relaxation and Rounding (Capital Budgeting)

(a) Binary IP formulation + IP solution

Decision variable:

$$x_j = \begin{cases} 1, & \text{if startup } j \text{ is funded} \\ 0, & \text{otherwise.} \end{cases}$$

Objective:

$$\max \sum_{j \in \text{STARTUPS}} r_j x_j$$

Budget constraints (Rounds 1–3):

$$\sum_j a_{1j} x_j \le 10, \quad \sum_j a_{2j} x_j \le 9, \quad \sum_j a_{3j} x_j \le 8$$

Additional constraints:

$$x_{S1} + x_{S3} \le 1, \qquad x_{S7} \le x_{S2}.$$

IP optimal solution (Gurobi/HiGHS):

$$x_{S3} = 1,\ x_{S4} = 1,\ x_{S5} = 1,\ \text{others } 0, \qquad z_{\text{IP}} = 47.$$

Budget usage: Round 1 = 7 (of 10), Round 2 = 8 (of 9), Round 3 = 7 (of 8).

(b) LP relaxation solution

LP relaxation: redeclare $x_j \in [0, 1]$.

LP optimal solution: $z_{\text{LP}} = 57$ with fractional values:

$$x_{S1} = 0.285714,\ x_{S2} = 0.857143,\ x_{S3} = 0.714286,\ x_{S8} = 0.785714,$$

and $x_{S5} = 1$ (others 0).

Startup	x_j (LP)
S1	0.285714
S2	0.857143
S3	0.714286
S4	0.000000
S5	1.000000
S6	0.000000
S7	0.000000
S8	0.785714

(c) Rounding and feasibility check

Rounding rule: $x_j \geq 0.5 \mapsto 1$, else 0 gives

$$\{S2, S3, S5, S8\} \text{ selected.}$$

Budget check: Round 1 usage $= 1 + 5 + 0 + 6 = 12 > 10$ (violates Round 1), and Round 3 usage $= 3 + 2 + 4 + 0 = 9 > 8$ (violates Round 3). So the rounded solution is **not feasible**.

(d) Integrality gap

$$\text{gap} = \frac{|z_{\text{IP}} - z_{\text{LP}}|}{|z_{\text{IP}}|} \times 100\% = \frac{|47 - 57|}{47} \times 100\% = 21.2766\%.$$

This is a **loose** relaxation.

Problem 2: Fixed-Charge Transportation

(a) MILP formulation

Decision variables:

$$x_{ij} \geq 0 \text{ (units shipped from plant } i \text{ to store } j), \quad y_{ij} \in \{0,1\} \text{ (route open).}$$

Objective:

$$\min \sum_i \sum_j c_{ij} x_{ij} + \sum_i \sum_j f_{ij} y_{ij}.$$

Supply:

$$\sum_j x_{ij} \leq \text{supply}_i \qquad \forall i.$$

Demand:

$$\sum_i x_{ij} \geq \text{demand}_j \qquad \forall j.$$

Switching/linking (big-M):

$$x_{ij} \leq M_{ij} y_{ij} \qquad \forall i, j, \quad M_{ij} = \min(\text{supply}_i, \text{demand}_j).$$

(b) LP relaxation vs IP + integrality gap

IP optimal objective: $z_{\text{IP}} = 1770$.

Open routes and shipments (IP):

Plant	Store	x_{ij}	c_{ij}	f_{ij}
A	Store 1	80	4	200
A	Store 3	40	1	150
B	Store 2	60	3	180
B	Store 4	40	2	120
C	Store 3	30	3	160
C	Store 5	50	2	150

LP relaxation objective: $z_{\text{LP}} = 1598.75$.

Nonzero LP flows and fractional route-opens:

Plant	Store	x_{ij} (LP)	y_{ij} (LP)
A	Store 1	50.00	0.625
A	Store 3	70.00	1.000
B	Store 2	60.00	1.000
B	Store 4	40.00	1.000
C	Store 1	30.00	0.375
C	Store 5	50.00	1.000

Integrality gap (minimisation):

$$\text{gap} = \frac{z_{\text{IP}} - z_{\text{LP}}}{z_{\text{IP}}} \times 100\% = \frac{1770 - 1598.75}{1770} \times 100\% = 9.6751\%.$$

Problem 3: Set Covering with Budget and Performance Tiers

Let $y_t \in \{0,1\}$ indicate whether tool t is selected.

(a) Minimum-cost covering

$$\min \sum_t \text{cost}_t y_t \quad \text{s.t.} \quad \sum_{t \in \mathcal{C}(i)} y_t \geq 1 \;\; \forall i \in \{1,\ldots,12\}.$$

Optimal solution: select $\{T3, T7, T8\}$, total cost = 140K.

(b) Budget-constrained max coverage (120K)

Let $z_i \in \{0,1\}$ indicate whether threat i is covered.

$$\max \sum_i z_i \quad \text{s.t.} \quad z_i \leq \sum_{t \in \mathcal{C}(i)} y_t \;\; \forall i, \qquad \sum_t \text{cost}_t y_t \leq 120.$$

Optimal solution: select $\{T3, T4, T5\}$, total cost = 120K, covers 11 threats (uncovered threat: 2).

(c) Redundancy: each threat covered by at least 2 tools

Replace covering constraints by:

$$\sum_{t \in \mathcal{C}(i)} y_t \geq 2 \qquad \forall i.$$

Optimal solution: select $\{T1, T2, T3, T4, T5, T6\}$, total cost = 255K.

(d) Reliability tiers: average reliability ≥ 2.5

Map: High = 3, Medium = 2, Low = 1. Add:

$$\frac{\sum_t \text{rel}_t y_t}{\sum_t y_t} \geq 2.5 \quad \Longleftrightarrow \quad \sum_t \text{rel}_t y_t \geq 2.5 \sum_t y_t.$$

Optimal solution: select $\{T3, T7, T8\}$, total cost = 140K (average reliability = $\frac{3+2+3}{3} = 2.6667$).

Problem 4: Emergency Shelter Location After a Natural Disaster

Variables:

$$y_j \in \{0,1\} \text{ (shelter } j \text{ open)}, \qquad x_{ij} \geq 0 \text{ (people from community } i \text{ assigned to shelter } j\text{)}.$$

Objective (in \$K; since \$100 per person-minute = 0.1K per person-minute):

$$\min \sum_j F_j y_j + 0.1 \sum_i \sum_j t_{ij} x_{ij}.$$

Demand balance:

$$\sum_j x_{ij} = \text{pop}_i \qquad \forall i.$$

Capacity:

$$\sum_i x_{ij} \leq \text{cap}_j y_j \qquad \forall j.$$

(b) Solution (open shelters, assignments, total cost)

Optimal shelters opened: all shelters $\{S1, \ldots, S8\}$.

Total cost: 4230K = 580K (fixed) +3650K (travel).

Shelter	Load	Capacity	Fixed cost (\$K)
S1	500	500	80
S2	400	400	65
S3	600	600	95
S4	180	350	55
S5	280	450	70
S6	390	550	90
S7	300	300	50
S8	400	500	75

Assignments (nonzero x_{ij}):

Community	Pop.	Assignment x_{ij} (travel min)
A1	200	S1: 90 (10); S8: 110 (15)
A2	150	S6: 150 (15)
A3	300	S3: 300 (10)
A4	180	S4: 180 (10)
A5	250	S5: 250 (10)
A6	120	S6: 120 (10)
A7	280	S5: 30 (15); S7: 250 (10)
A8	160	S8: 160 (10)
A9	350	S2: 230 (15); S6: 120 (20)
A10	90	S3: 40 (15); S7: 50 (15)

Community	Pop.	Assignment x_{ij} (travel min)
A11	220	S1: 220 (15)
A12	170	S2: 170 (10)
A13	130	S8: 130 (15)
A14	260	S3: 260 (10)
A15	190	S1: 190 (10)

(c) Equity constraint: travel time ≤ 30 minutes

Equity enforcement: for any (i, j) with $t_{ij} > 30$, force $x_{ij} = 0$.

Result: solution unchanged; cost remains 4230K.

(d) Redundancy constraint: 2 open shelters within 40 minutes

For each community i:

$$\sum_{j:t_{ij}\leq 40} y_j \geq 2.$$

Result: solution unchanged; cost remains 4230K.

Appendix: AMPL (.mod/.dat) listings (same filenames as notebook)

Problem 1: `vc_funding.mod`

```
set STARTUPS;

param r1 {STARTUPS} >= 0;
param r2 {STARTUPS} >= 0;
param r3 {STARTUPS} >= 0;
param ret {STARTUPS};

param B1 >= 0;
param B2 >= 0;
param B3 >= 0;

var x {STARTUPS} binary;

maximize TotalReturn: sum {j in STARTUPS} ret[j] * x[j];

subject to Budget1: sum {j in STARTUPS} r1[j]*x[j] <= B1;
subject to Budget2: sum {j in STARTUPS} r2[j]*x[j] <= B2;
subject to Budget3: sum {j in STARTUPS} r3[j]*x[j] <= B3;

# S1 and S3: at most one
subject to Compete: x["S1"] + x["S3"] <= 1;

# S7 requires S2
subject to Require: x["S7"] <= x["S2"];
```

Problem 1: vc_funding.dat

```
set STARTUPS := S1 S2 S3 S4 S5 S6 S7 S8;

param: r1 r2 r3 ret :=
S1 3 2 0 12
S2 1 4 3 15
S3 5 0 2 18
S4 2 3 1  9
S5 0 5 4 20
S6 4 1 3 14
S7 2 2 5 16
S8 6 0 0 10
;

param B1 := 10;
param B2 := 9;
param B3 := 8;
```

Problem 2: transport_fc.mod

```
set PLANTS;
set STORES;

param supply {PLANTS} >= 0;
param demand {STORES} >= 0;

param c {PLANTS, STORES} >= 0;   # variable cost
param f {PLANTS, STORES} >= 0;   # fixed cost

param M {i in PLANTS, j in STORES} := if supply[i] < demand[j] then supply[i] else demand[j];

var x {PLANTS, STORES} >= 0;
var y {PLANTS, STORES} binary;

minimize TotalCost:
  sum {i in PLANTS, j in STORES} c[i,j]*x[i,j] + f[i,j]*y[i,j];

subject to Supply {i in PLANTS}:
  sum {j in STORES} x[i,j] <= supply[i];

subject to Demand {j in STORES}:
  sum {i in PLANTS} x[i,j] >= demand[j];

subject to Link {i in PLANTS, j in STORES}:
  x[i,j] <= M[i,j]*y[i,j];
```

Problem 2: transport_fc.dat

```
set PLANTS := A B C;
set STORES := 1 2 3 4 5;

param supply :=
A 120
B 150
```

```
C 100
;

param demand :=
1 80
2 60
3 70
4 40
5 50
;

param c : 1 2 3 4 5 :=
A 4 8 1 5 7
B 7 3 6 2 4
C 5 6 3 7 2
;

param f : 1 2 3 4 5 :=
A 200 350 150 250 300
B 300 180 280 120 200
C 250 270 160 320 150
;
```

Problem 3: cybersecurity.mod

```
set THREATS;
set TOOLS;

param cost {TOOLS} >= 0;
param rel  {TOOLS} >= 0;         # High=3, Medium=2, Low=1
param cover {THREATS, TOOLS} binary;

var y {TOOLS} binary;

minimize TotalCost: sum {t in TOOLS} cost[t]*y[t];

subject to CoverAll {i in THREATS}:
  sum {t in TOOLS} cover[i,t]*y[t] >= 1;

# For part (c): change RHS to 2
# For part (d): add avg reliability constraint:
#   sum rel[t]*y[t] >= 2.5 * sum y[t];
```

Problem 3: cybersecurity.dat

```
set THREATS := 1 2 3 4 5 6 7 8 9 10 11 12;
set TOOLS := T1 T2 T3 T4 T5 T6 T7 T8;

param cost :=
T1 45
T2 30
T3 55
T4 25
T5 40
T6 60
```

```
T7 35
T8 50
;

param rel :=
T1 3
T2 2
T3 3
T4 1
T5 2
T6 3
T7 2
T8 3
;

param cover (tr):
      T1 T2 T3 T4 T5 T6 T7 T8 :=
  1    1  0  0  1  0  0  0  1
  2    1  1  0  0  0  0  1  0
  3    0  1  1  0  0  0  0  1
  4    0  0  1  0  1  0  0  0
  5    1  0  0  1  0  0  1  0
  6    0  1  0  0  1  0  0  1
  7    0  0  1  0  0  1  0  0
  8    1  0  0  0  1  0  1  0
  9    0  0  1  0  0  1  0  1
 10    0  0  0  1  0  1  1  0
 11    0  0  0  0  1  1  0  1
 12    0  0  1  0  0  1  0  0
;
```

Problem 4: `shelter.mod`

```
set COMM;
set SHEL;

param pop   {COMM} >= 0;
param cap   {SHEL} >= 0;
param F     {SHEL} >= 0;          # fixed cost ($K)
param t     {COMM, SHEL} >= 0;    # travel time (min)

param pen := 0.1;                 # $100 per person-min = 0.1K

var x {COMM, SHEL} >= 0;          # people assigned
var y {SHEL} binary;              # shelter open

minimize TotalCost:
  sum {j in SHEL} F[j]*y[j] + pen * sum {i in COMM, j in SHEL} t[i,j]*x[i,j];

subject to Assign {i in COMM}:
  sum {j in SHEL} x[i,j] = pop[i];

subject to Capacity {j in SHEL}:
  sum {i in COMM} x[i,j] <= cap[j]*y[j];

# Part (c) equity: enforce x[i,j]=0 when t[i,j] > 30
# subject to Equity {i in COMM, j in SHEL: t[i,j] > 30}:
```

```
#   x[i,j] = 0;

# Part (d) redundancy: at least 2 open shelters within 40 min
# subject to Redundant {i in COMM}:
#   sum {j in SHEL: t[i,j] <= 40} y[j] >= 2;
```

Problem 4: `shelter.dat`

```
set COMM := A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15;
set SHEL := S1 S2 S3 S4 S5 S6 S7 S8;

param pop :=
A1 200
A2 150
A3 300
A4 180
A5 250
A6 120
A7 280
A8 160
A9 350
A10 90
A11 220
A12 170
A13 130
A14 260
A15 190
;

param cap :=
S1 500
S2 400
S3 600
S4 350
S5 450
S6 550
S7 300
S8 500
;

param F :=
S1 80
S2 65
S3 95
S4 55
S5 70
S6 90
S7 50
S8 75
;

param t (tr):
      S1 S2 S3 S4 S5 S6 S7 S8 :=
A1    10 25 30 45 20 35 50 15
A2    30 10 20 35 40 15 25 45
A3    20 35 10 15 30 40 20 25
A4    40 15 25 10 35 20 30 50
```

```
A5     15 40 35 30 10 25 45 20
A6     35 20 15 40 25 10 30 35
A7     25 30 40 20 15 35 10 30
A8     45 25 20 30 35 15 40 10
A9     20 15 30 25 40 20 35 30
A10    30 40 15 20 25 30 15 40
A11    15 35 25 30 20 40 25 20
A12    40 10 35 25 30 20 35 25
A13    25 20 40 35 15 25 30 15
A14    35 30 10 15 25 30 40 35
A15    10 25 20 40 35 15 20 30
;
```